

Where,

$$W_{NS}(\alpha) = \frac{2(\pi\mu\gamma(\frac{bQ}{2}))^{-\frac{Q-2\alpha}{2b}} \pi(\alpha - Q/2)}{\Gamma(1 + b(\alpha - Q/2))\Gamma(1 + \frac{1}{b}(\alpha - Q/2))} \quad (\text{B.8})$$

- $\Gamma_{NS}(x)$ has a pole at 0, the behaviour near the pole is given by

$$\Gamma_{NS}(x) \xrightarrow{x \rightarrow 0} \frac{\Gamma_{NS}(Q)}{\pi x}, \quad \Upsilon'_{NS}(0) = \frac{\pi}{\Gamma_{NS}^2(Q)}. \quad (\text{B.9})$$

- **Upsilon functions:**

$$\frac{\Upsilon_{NS}(2x)}{\Upsilon_{NS}(2x - Q)} = \mathcal{G}_{NS}(x) \lambda^{-\frac{Q-2x}{b}}, \quad (\text{B.10})$$

where,

$$\mathcal{G}_{NS}(x) = \frac{W_{NS}(Q - x)}{W_{NS}(x)}. \quad (\text{B.11})$$

C Torus two-point function (Liouville CFT) with general left and right moving temperature

We first calculate the torus two-point function in 2d CFT where we have some intermediate left and right moving temperature β_L and β_R , respectively. We insert the two operators $\mathcal{O}_1$ and $\mathcal{O}_2$ in the *Euclidean time circle*. To distinguish them from the insertion in the spatial circle, we use a *tilde* symbol. Now we use the modular crossing kernels and Virasoro fusion kernel to calculate the torus two-point function in the following fashion Fig. (4), considering $\alpha = \frac{Q}{2} + iP$ (where P is the Liouville Momenta),

$$\begin{aligned} \mathcal{F}_{hh'}^{\tilde{N}} &= \widetilde{\mathcal{F}_{P'P}^{OPE}}, \\ &= \int \frac{dP_2}{2} \frac{dP_1}{2} \mathbb{S}_{P'P_2}[P] \mathbb{F}_{P,P_1} \begin{bmatrix} \tilde{P} & P_2 \\ \tilde{P} & P_2 \end{bmatrix} \mathcal{F}_{P_1,P_2}^N. \end{aligned} \quad (\text{C.1})$$

Now we know that the fusion kernel can be written in the following integral form [133],

$$\mathbb{F}_{P'',P} \begin{bmatrix} P^* & \tilde{P} \\ P^* & \tilde{P} \end{bmatrix} = P(\alpha_i; \alpha'', \alpha) P(\alpha_i; Q - \alpha'', Q - \alpha) \int_{\mathcal{C}'} \frac{ds}{i} \prod_{k=1}^4 \frac{\mathcal{S}_b(s + U_k)}{\mathcal{S}_b(s + V_k)} \quad (\text{C.2})$$

where,

$$P(\alpha_i; \alpha'', \alpha) = \frac{\Gamma_b^2(\alpha'' + \alpha^* - \tilde{\alpha}) \Gamma(\alpha'' + Q - \tilde{\alpha} - \alpha^*) \Gamma(\alpha'' - Q + \tilde{\alpha} + \alpha^*) \Gamma_b(2\alpha)}{\Gamma_b^2(\alpha) \Gamma(\alpha + Q - 2\tilde{\alpha}) \Gamma(\alpha - Q + 2\alpha^*) \Gamma_b(2\alpha - Q)} \quad (\text{C.3})$$

along with,

$$\mathcal{S}_b(x) := \frac{\Gamma_b(x)}{\Gamma_b(Q - x)}. \quad (\text{C.4})$$